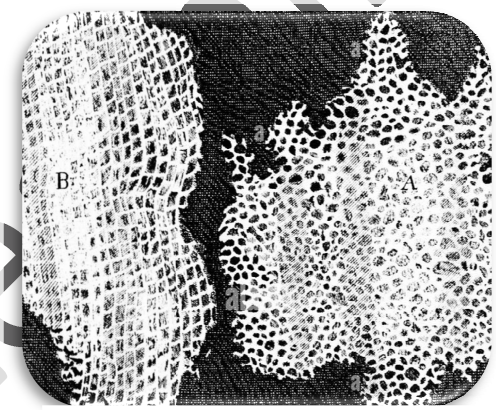
CELL: THE FUNDAMENTAL UNIT OF LIFE

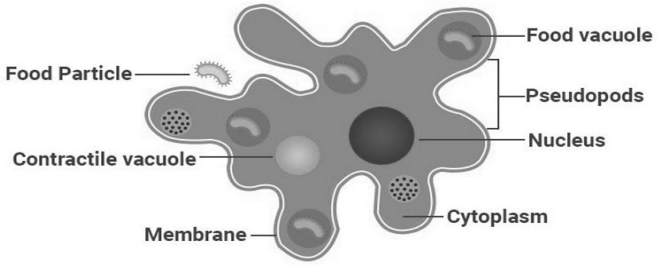
- All organisms that we observe around are made up of cells.
- Cells are called as basic functional unit of life.
- There are also single cells that live on their own.
- Cells perform functions like respiration, obtaining nutrition, and clearing of waste material, or forming new proteins
- Cell is a Latin word for 'a little room'.
- **History of Cell Discovery:**
  - ✓ Cells were first discovered by **Robert Hooke** in **1665**.
  - ✓ Observations of Robert Hooke in 1665:
    - Robert Hooke examined the thin slice of cork(Cork is a substance which comes from bark of tree).
    - He observed cork with a self-designed microscope. It was a primitive microscope.
    - He saw that the cork resembled the structure of a honeycomb consisting of many little compartments.
    - Robert Hooke called these boxes cells.
    - This was the very first time that someone had observed that living things appear to consist of separate units.



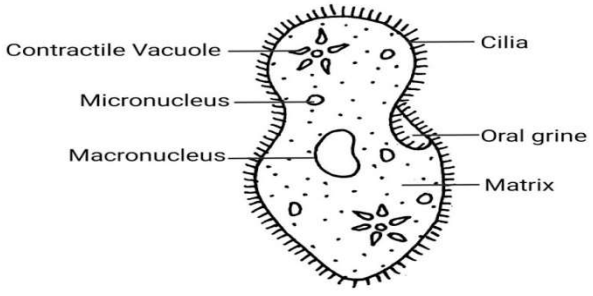
Picture of cells seen by Robert Hooke

  - ✓ **Antonie van Leeuwenhoek (1674),** with the improved microscope, discovered the free living cells in pond water for the first time.
  - ✓ **Robert Brown** in **1831** discovered the nucleus in the cell.
  - ✓ **Jan Evangelista Purkinje in 1839** coined the term '**protoplasm**' for the fluid substance of the cell.
  - ✓ **Matthias Jakob Schleiden(1838**)and **Theodor Schwann(1839**)presented **Cell theory** and stated:
    1) All the plants and animals are composed of cells
    2) The cell is the basic unit of life
  - ✓ **Rudolf Virchow(1855) expended the cell theory by suggesting that:**
    3) All cells arise from pre-existing cells.
  - ✓ The complex structure of the cell and its various organelles are possible to observe by Electron microscope which was discovered in 1940.
- **Types of Organism as per their cell constituent:**

a) **<u>Unicellular</u>**: An Organism made up of Single Cell only is called as Uni-Cellular. For Example: Amoeba, Chlaymydomonas, Paramoecium and Bacteria.



Amoeba



Paramecium

Conceptual Learning

b) **Multi- cellular:** An Organism made up of many cells and groups of cells is called as Multicellular Organism. For Example: Fungi, Plants and Animals.

- Every multi-cellular organism has come from a single cell. Cells divide to produce cells of their own kind. Thus, all cells come from pre-existing cells.
- Shape and Size of Cell:
  - The shape and size of cells depends on the specific function they perform.
  - Some cells have changing shapes like Amoeba.
  - In Some cells shape could be more or less fixed and peculiar. For example: Nerve cells have a typical shape.
- **Functioning of Cell:**
  - ✓ Each living cell has the capacity to perform certain basic functions that are characteristic of all living forms.
  - ✓ Division of work and function is seen within a single cell.
  - ✓ Specific components of a cell to perform special functions within it are known as **Cell Organelles.**
  - ✓ Functions of Cell Organelles include making new material in the cell, clearing up the waste material from the cell etc.
  - ✓ A cell is able to live and perform all its functions because of these organelles.
  - ✓ All cells are found to have almost same organelles, irrespective of their function and organism they are found in.

- **Structural Organization of a Cell:** All cells have three same features as their structure:
  i. Plasma Membrane Or Cell Membrane
  ii. Cytoplasm
  iii. Nucleus

❖ Plasma Membrane or Cell Membrane

- ✓ Plasma Membrane is the outermost covering of the cell that separates the contents of the cell from its external environment.
- ✓ The plasma membrane is flexible and is made up of organic molecules called lipids and proteins.
- ✓ Its structure can be seen by electron microscope only.
- ✓ The plasma membrane allows or permits the entry and exit of some materials in and out of the cell.
- ✓ As cell membrane prevents and allows movement of materials in and out of cell, it is called a **Selectively Permeable Membrane.**
- ✓ Spontaneous movement of a substance from a region of high concentration to a region where its concentration is low is called **Diffusion**. Substances like carbon dioxide or oxygen can move across the cell membrane by the process of Diffusion.
- ✓ The movement of water molecules through a selectively permeable membrane is called **Osmosis.**
- ✓ The movement of water across the plasma membrane is also affected by the amount of substance dissolved in water. Osmosis is the net diffusion of water across a selectively permeable membrane toward a higher solute concentration.

Conceptual Learning

- ✓ If a cell is put inside a solution, water moves outside and inside from semi permeable membrane of cell. On the basis of, concentration of water inside and outside of a cell, we can categorize it in three categories:
  i. If the medium surrounding the cell has a higher water concentration than the cell, or if is very dilute, the cell will gain water by osmosis. Such a solution is known as a **Hypotonic solution**. In this case cell is likely to swell up.
  ii. If the medium has exactly the same water concentration as the cell, there will be no net movement of water across the cell membrane. Such a solution is known as an **Isotonic solution**. The cell will stay the same size.
  iii. If the medium has a lower concentration of water than the cell, meaning that it is a very concentrated solution, the cell will lose water by osmosis. Such a solution is known as a **hypertonic solution**. Therefore the cell will shrink.
- ✓ **Examples of Osmosis:**
  i. If a de-shelled egg or Raisin is put in water, it will swell up and if it sis put in a concentrated solution it will shrink.
  ii. Absorption of water by plant roots is also an example of osmosis.
- ✓ Advantages of Plasma membrane:
  - It helps in exchange of gases, water and nutrients in cell.
  - Due to flexibility of cell membrane, a few cells are able to engulf food from environment. This processes are known as **Endocytosis. Amoeba** acquires its food through such Endocytosis.

➢ **Cell Wall:**

- ✓ A rigid outer covering outside plant cell other than plasma membrane is called cell wall.
- ✓ The plant cell wall is mainly composed of cellulose. Cellulose is a complex substance and provides structural strength to plants.
- ✓ When a living plant cell loses water through osmosis there is shrinkage or contraction of the contents of the cell away from the cell wall. This phenomenon is known as **Plasmolysis.**
- ✓ It is freely permeable membrane.
- ✓ Cell walls permit the cells of plants, fungi and bacteria to withstand very dilute (hypotonic) external media without bursting. The cell swells, building up pressure against the cell wall. The wall exerts an equal pressure against the swollen cell. Because of their walls, such cells can withstand much greater changes in the surrounding medium than animal cells.

<u>Note:</u> Different According to their chemical composition different regions of cells get colored differentially. Some regions appear darker than other regions. Iodine solution or saffranin solution or methylene blue can be used to view colors.

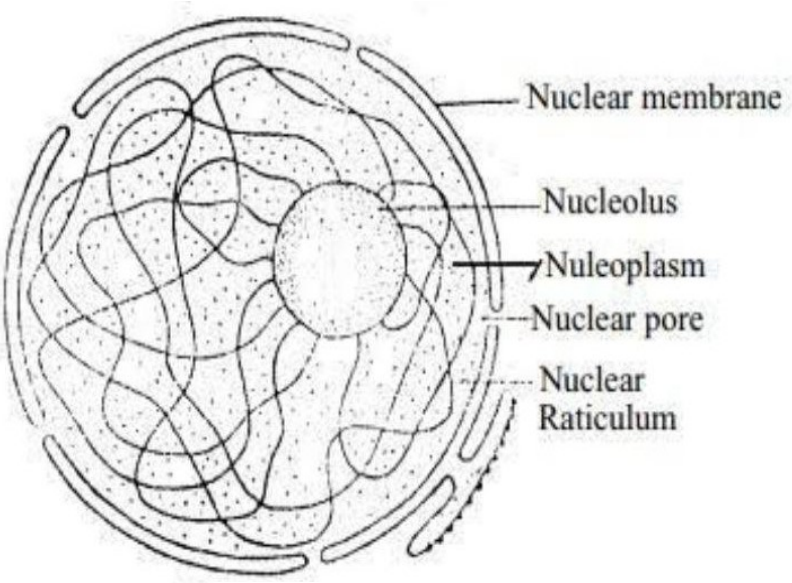
❖ Nucleus

➢ **Parts and sturucture:**

- ✓ The nucleus has a double layered covering called nuclear membrane.
- ✓ The nuclear membrane has pores which allow the transfer of material from inside the nucleus to its cytoplasm.

Conceptual Learning

- ✓ The nucleus contains chromosomes, which are visible as rod-shaped structures only when the cell is about to divide.
- ✓ Chromosomes contain information for inheritance of characters from parents to next generation in the form of DNA (**Deoxyribo Nucleic Acid**) molecules.
- ✓ Chromosomes are composed of DNA and protein.
- ✓ DNA molecules contain the information necessary for constructing and organizing cells.



Nucleus

- ✓ Functional segments of DNA are called **Genes.**
- ✓ In a cell which is not dividing, this DNA is present as part of chromatin material. Chromatin material is visible as entangled mass of thread like structures.
- ✓ Whenever the cell is about to divide, the chromatin material gets organized into chromosomes.

➢ **Functions of Nucleus**

- ✓ The nucleus plays a central role in cellular reproduction. **Cellular Reproduction** is the process by which a single cell divides and forms two new cells.
- ✓ Nucleus also plays a crucial part, along with the environment, in determining the way the cell will develop and what form it will exhibit at maturity, by directing the chemical activities of the cell.

➢ **Types of Cells:** On the basis of nuclear region, Cell is divided into two types:

i. **Prokaryotes:** Organisms which are not having a defined nuclear region due to absence of nuclear membrane. Examples are Bacteria and Archea.

ii. **Eukaryotes:** Organisms which are having a defined nuclear region due to presence of nuclear membrane. Fungi, Algae, Amoeba, Euglena, Plasmodium, Animals, and Plants

Prokaryotes	Eukaryotes
Nuclear membrane in cells is absents	Nuclear membrane in cells is present.
Chlorophyll is associated with membranous vesicles.	Chlorophyll is associated with plastids.
Generally unicellular	Unicellular as well as multicellular.
Comparatively smaller.	Comparatively Larger.
Lysosomes are absent.	Lysosomes are present.
Mitochondria and Endoplasmic Reticulam are absent.	Mitochondia and Endoplasmic Reticulam are present.
Membrane-bound organelles are absent.	Membrane-bound organelles are present.
Many functions of cell organelles are performed by poorly organized parts of cytoplasm.	Many functions of cell organelles are performed by poorly organized parts of cell organelles themselves.

Conceptual Learning

Note: The nucleoid (meaning nucleus-like) is an irregularly shaped region within the prokaryotic cell that contains all or most of the genetic material.

❖ Cytoplasm

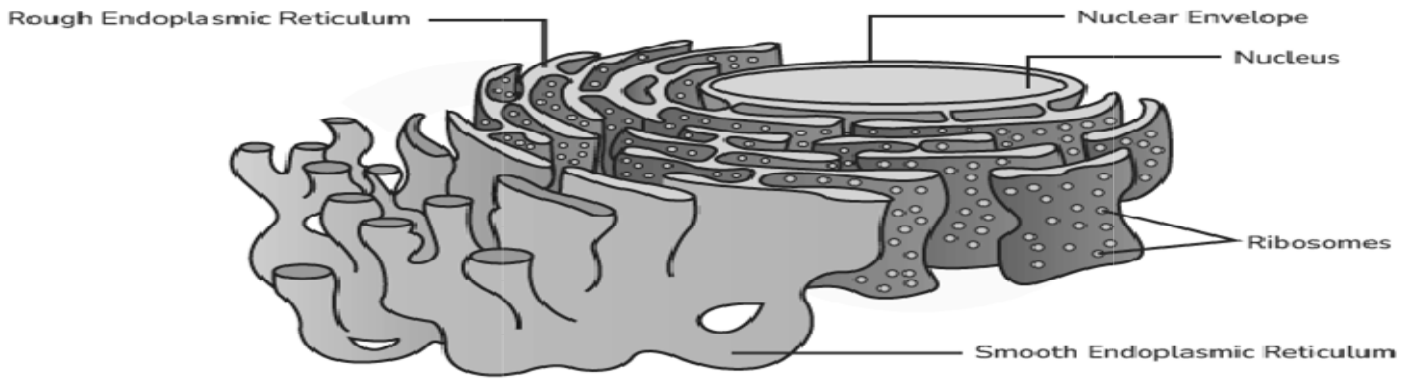
- ✓ A large region of fluid content of cell enclosed by the cell membrane is called **Cytoplasm.**
- ✓ When colored by Iodine or saffranin, cytoplasm takes up very little stain.

➢ **Cell Organelles:**

- ✓ Cytoplasm contains many specialized cell organelles.
- ✓ Each of these organelles performs a specific function for the cell.
- ✓ Some of these organelles are visible only with an electron microscope.
- ✓ Cell organelles are enclosed by membranes in Eukaryotes whereas membrane bound cell organelles are absent in Prokaryotes.
- ✓ Large and complex cells, including cells from multicellular organisms, need a lot of chemical activities to support their complicated structure and function. To keep these activities of different kinds separate from each other, these cells use membrane-bound organelles within themselves.
- ✓ Significance of Membranes illustrated by virus: Viruses lack any membranes and hence do not show characteristics of life until they enter a living body and use its cell machinery to multiply.
- ✓ Various Cell organelles are:
  - (i) Endoplasmic reticulam
  - (ii) Golgi Appratus
  - (iii) Lysosomes
  - (iv) Mitochondria
  - (v) Plastids
  - (vi) Vacuoles

➢ **Endoplasmic Reticulam**

- **Structure:**



- ✓ The endoplasmic reticulum (ER) is a large network of membrane-bound tubes and sheets.
- ✓ It looks like long tubules or round or oblong bags (vesicles).
- ✓ The ER membrane is similar in structure to the plasma membrane.
- ✓ Although the ER varies greatly in appearance in different cells, it always forms a network system.

- **Types of ER and their function:**

(i) Rough endoplasmic reticulum (RER)

✓ RER looks rough under a microscope because it has particles called Ribosomes attached to its surface. The Ribosomes are present in all active cells, are the sites of protein manufacture.

✓ The manufactured proteins are then sent to various places in the cell depending on need, using the ER.

(ii) Smooth endoplasmic reticulum (SER).

✓ The SER helps in the manufacture of fat molecules, or lipids, which important for cell function.

- **Function of ER:**

✓ ER helps in manufacture of proteins and fats.

✓ ER serves as channels for the transport of materials (especially proteins) between various regions of the cytoplasm or between the cytoplasm and the nucleus.

✓ The ER also functions as a cytoplasmic framework providing a surface for some of the biochemical activities of the cell. In the liver cells of the group of animals called vertebrates, SER plays a crucial role in detoxifying many poisons and drugs.

- **Function of Proteins and Lipids manufactured in ER:**

✓ The process by which some proteins and lipids(Manufactured in ER) helps in building the cell membrane is known as Membrane Biogenesis.

✓ Some other proteins and lipids function as enzymes and hormones.

➢ **The Golgi Apparatus:** First described by Camillo Golgi

- **About Camillo Golgi**

✓ He was chief medical officer at the Hospital for the chronically sick at Abbiategrasso(Italy) and was researching in Nervous System.

✓ Golgi carried out a method of staining individual nerve and cell structures which is known as Black Reaction. This method uses a weak solution of silver nitrate and is particularly valuable in tracing the processes and most delicate ramifications of cells.

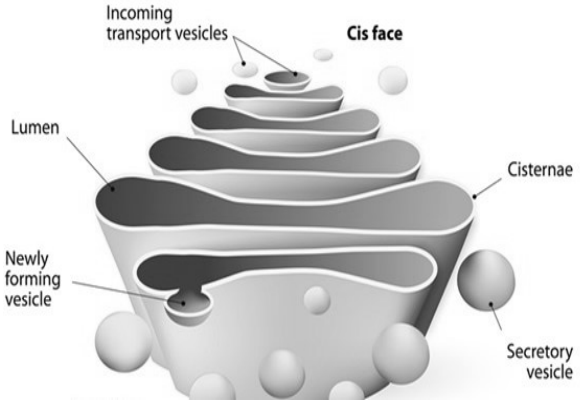
✓ He shared the Nobel prize in 1906 with Santiago Ramony Cajal for their work on the structure of the nervous system.

- **Structure of Golgi apparatus**

✓ Consists of a system of membrane-bound vesicles (flattened sacs) arranged approximately parallel to each other in stacks.

✓ These membrane bound vesicles are called Cisterns.

✓ These membranes often have connections with the membranes of ER and therefore constitute another portion of a complex cellular membrane system.



- **Function of Golgi Apparatus**

✓ The material synthesized near the ER is packaged and dispatched to various targets inside and outside the cell through the Golgi apparatus.

Conceptual Learning

✓ Its functions include the storage, modification and packaging of products in vesicles.

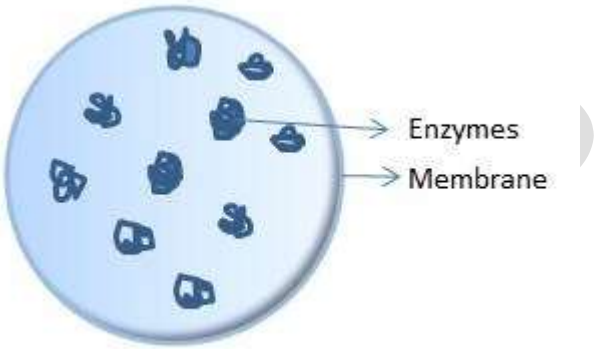
✓ In some cases, complex sugars may be made from simple sugars in the Golgi apparatus.

✓ The Golgi apparatus is also involved in the formation of Lysosomes.

➢ **Lysosomes**

- **Structure of Golgi apparatus:**

✓ Lysosomes are membrane-bound sacs filled with digestive enzymes. These enzymes are made by RER.



- **Function of Lysosomes:**

✓ Lysosomes are a kind of waste disposal system of the cell.

✓ These help to keep the cell clean by digesting any foreign material(bacteria or food) as well as worn-out cell organelles. Lysosomes break complex substances into simpler substances with the help of powerful digestive enzymes.

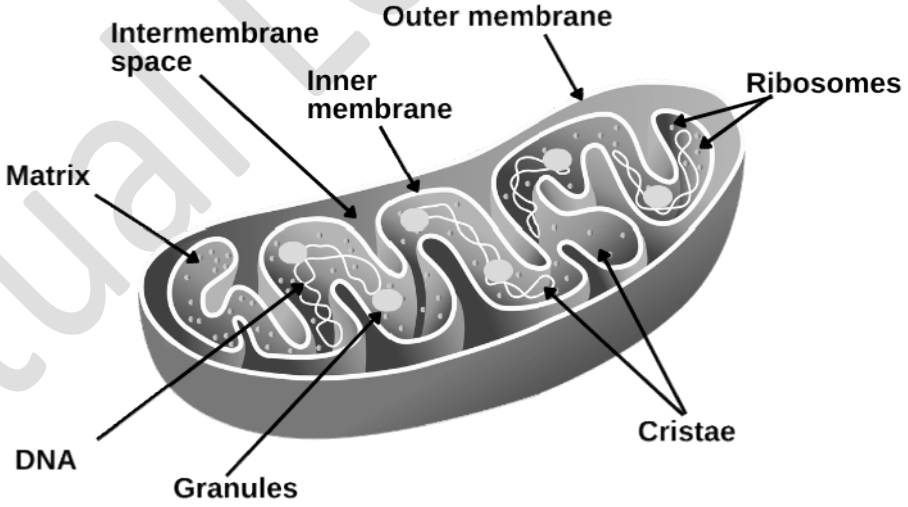
✓ During the disturbance in cellular metabolism, lysosomes may burst and the enzymes digest their own cell. Therefore, **lysosomes are also known as the 'suicide bags'**

➢ **Mitochondria:** Power house of cell and site for ATP generation.

- **Structure of Mitochondria**

✓ Mitochondriya is double membranes cell organelle.

✓ The outer membrane is porous while the inner membrane is deeply folded. These folds increase surface area for ATP generating chemical reactions.



- **Function of Mitochondria:** The energy required for various chemical activities needed for life is released by mitochondria in the form of ATP (Adenosine triphopshate) molecules. **ATP is known as the energy currency of the cell.**
- **Mitochondria are strange organelles in the sense that they have their own DNA and ribosomes.** Therefore, mitochondria are able to make some of their own proteins.

➢ **Plastids**

✓ Plastid is a double membrane-bound organelle involved in the synthesis and storage of food, commonly found within the cells of photosynthetic plants.

✓ Like the mitochondria, plastids also have their own DNA and ribosomes.

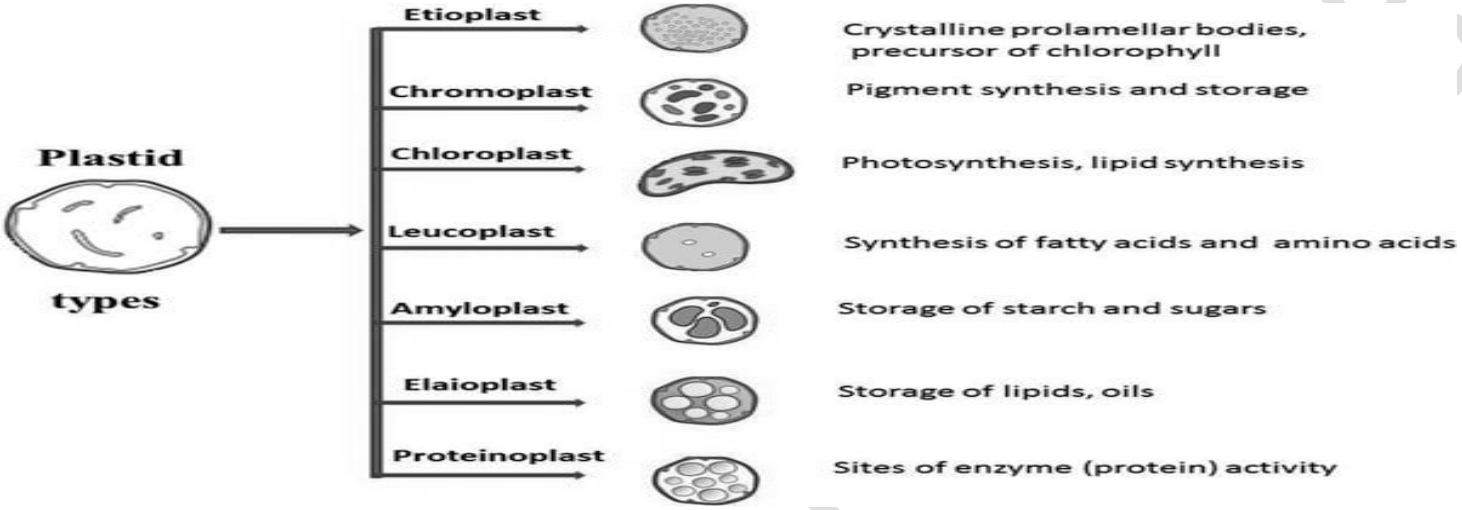
✓ Three important types of plastids:

(i) **Chromoplasts** (coloured plastids): These plastids have coloured pigments which gives colors to flowers and leaves that is good for attacting insects which encourages pollination.

Conceptual Learning

(ii) **Chloroplast**(Plastids with Chlorophyll): Chromoplasts containing the pigment chlorophyll are known as chloroplasts. Chloroplasts are important for photosynthesis in plants. The internal organisation of the Chloroplast consists of numerous membrane layers embedded in a material called the stroma. These are similar to mitochondria in external structure.

(iii) **Leucoplasts** (white or colourless plastids): Leucoplasts are primarily organelles in which materials such as starch, oils and protein granules are stored.

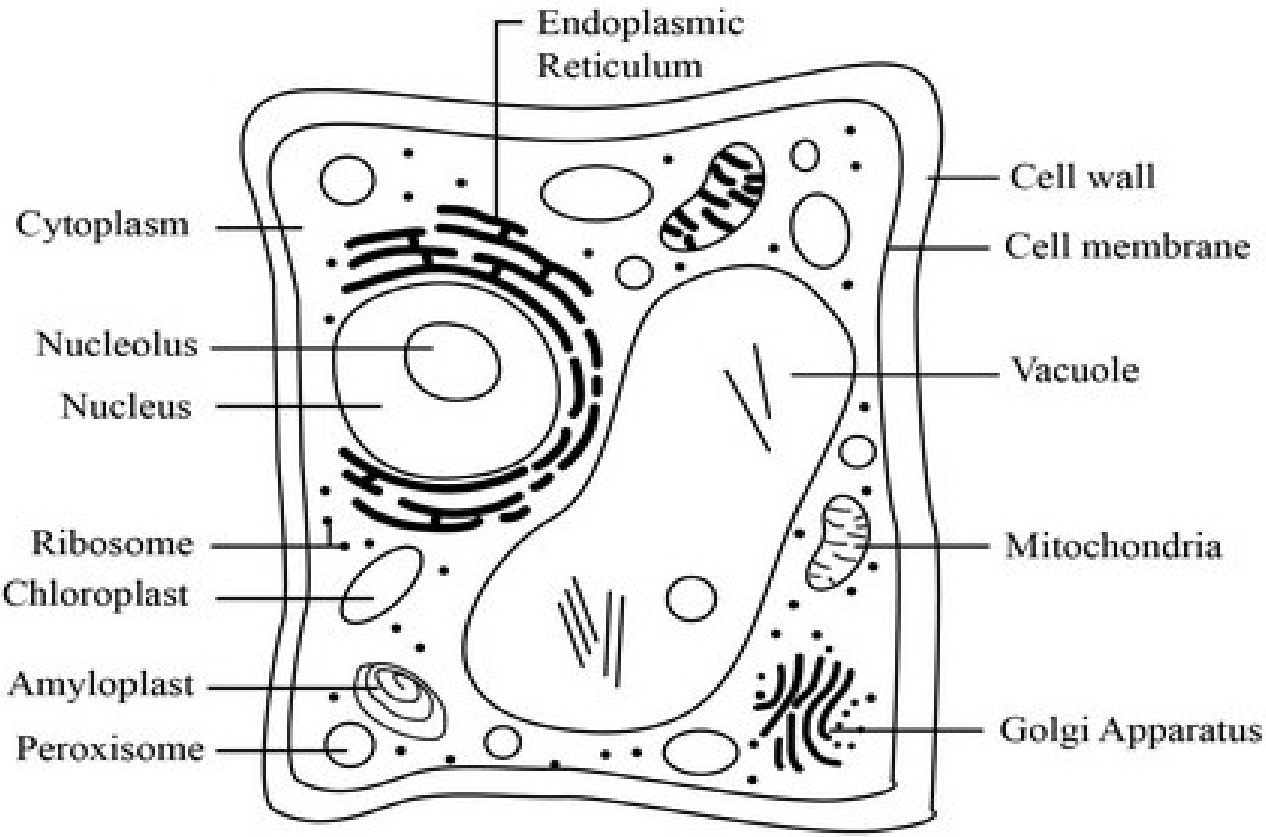


- **Vacuoles**
  - ✓ Vacuoles are storage sacs for solid or liquid contents.
  - ✓ Vacuoles are small sized in animal cells while plant cells have very large vacuoles.
  - ✓ The central vacuole of some plant cells may occupy 50-90% of the cell volume.
  - ✓ In plant cells vacuoles are full of cell sap and provide turgidity(swelled up) and rigidity to the cell.
  - ✓ Many substances of importance in the life of the plant cell are stored in vacuoles. These include amino acids, sugars, various organic acids and some proteins.
  - ✓ In single-celled organisms like Amoeba, the food vacuole contains the food items that the Amoeba has consumed.
  - ✓ In some unicellular organisms, specialized vacuoles also play important roles in expelling excess water and some wastes from the cell.
  - ✓ The **Tonoplast** is the membrane that surrounds the central or large vacuole of the plant.

❖ **Difference between Plant Cell and Animal Cell**

Plant Cell	Animal Cell
Cell Wall is present.	Cell Wall is absent.
Few large or a single, centrally positioned vacuole	Usually small and numerous
Nucleus lies on one side of the cell	Nucleus lies on centre of the cell
Plastids are present.	Plastids are absent.
Cillia is absent.	Cilia is present in most of animal cells.
Mitochondria are few.	Mitochondria are in large numbers.
Autotrophic mode of nutrition.	Hetrotrophic mode of nutrition.

PLANT CELL

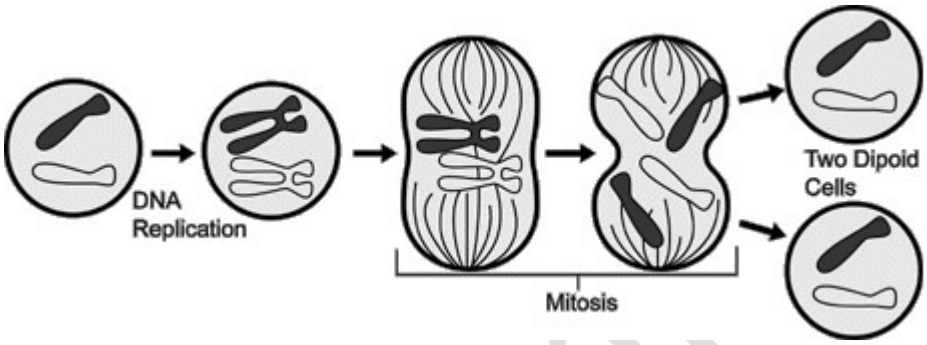


❖ **Cell Division:**

✓ The process by which new cells are made is called cell division.

✓ New cells are formed in organisms in order to grow, to replace old, dead and injured cells, and to form gametes required for reproduction.

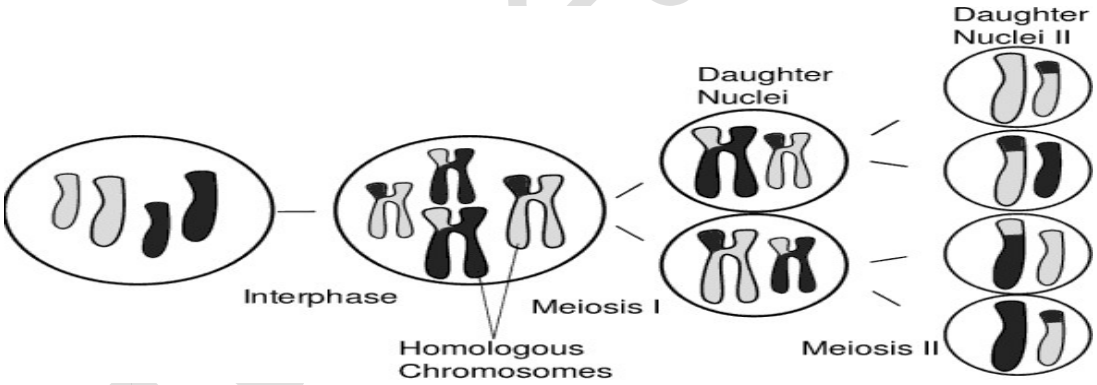
✓ There are two main types of cell division:



(i) Mitosis:

- Most of the cells divide for growth by this process.
- In this process, each cell called mother cell divides to form two identical daughter cells.
- The daughter cells have the same number of chromosomes as mother cell.
- It helps in growth and repair of tissues in organisms.

(ii)Meiosis:



- Meiosis is a process of cell division which includes two consecutive divisions
- Specific cells of reproductive organs or tissues in animals and plants divide to form gametes, which after fertilization give rise to offspring.
- When a cell divides by meiosis it produces four new cells instead of two.
- The new cells only have half the number of chromosomes than that of the mother cells.

Mitosis	Meiosis
It is the cell division that forms two daughter cells each with the same number and type of chromosomes as the parent cell.	It is the cell division which forms four daughter cells each with half the number of chromosomes as the parent cell.
It is happens asexual type of reproduction.	It happens in sexual mode of reproduction.
Two divisions of nucleus.	Four divisions of nucleus.
Number of chromosome remains the same.	Number of chromosome remains the half.
Function is General growth and repair and reproduction of cells.	Function is to form gametes which undergoes sexual reproduction

❖ **Some Extra Definitions:**

- ✓ **Contractile Vacuole** is a specialized type of vacuole that regulates the quantity of water inside a cell.
- ✓ **Flagella** are the organelles for bacterial locomotion. These supramolecular structures extend from the cytoplasm to the cell exterior.
- ✓ **Cilia:** are small, slender, hair-like structures present on the surface of all mammalian cells.
- ✓ **Cilliates** are single-celled organisms that, at some stage in their life cycle, possess cilia, short hairlike organelles used for locomotion and food gathering.
- ✓ **Macronucleus** controls the vegetative functions and the **Micronucleu**s controls the reproductive functions. Certain ciliates have two nucleus.
- ✓ **Nucleolus is** the nucleolus is a spherical structure found in the cell's nucleus whose primary function is to produce and assemble the cell's ribosomes.
- ✓ **Neucleoplasm/Karyoplasm** is the contents of the cell nucleus.
- ✓ **Nuclear Reticullam** the diffuse intermeshed granular threads that represent the chromosomes in the resting nucleus.
- ✓ **Nuclear Envelope** is the nuclear Membrane.
- ✓ **Cis face** and **Trans Face** are the two faces of Golgi apparatus. The cis face of a Golgi stack is the end of the organelle where substances enter from the endoplasmic reticulum for processing, while the trans face is where they exit in the form of smaller detached vesicles.
- ✓ **Secretory vesicles** consist of the material that has to be excreted from the cell
- ✓ **Granuales** in cells are tiny grain like particles that may be hard to view under a light microscope. Most often the term granule is used to describe secretory vesicles.
- ✓ **Lumen** is the space within the tubes, tracts, cavities, and cells of the body.
- ✓ Pinocytic vesicles or pinocytotic vesicles or fluid-filled vesicles, are small membrane-bound sacs found in the cytoplasm of cells. They are formed through a process called pinocytosis, which is a type of endocytosis where the cell takes in small droplets of extracellular fluid along with the substances dissolved in it. The function of pinocytic vesicles is to transport and internalize extracellular fluids and the substances dissolved in them into the cell.
- ✓ **Centrioles** are cell organelles All animal cells have two centrioles. They help the cell during cell division. They work during the process of mitosis and meiosis.
- ✓ **Peroxysome** are small vesicles, single membrane-bound organelles found in the eukaryotic cells. They contain digestive enzymes for breaking down toxic materials in the cell and oxidative enzymes for metabolic activity.
- ✓ **Dictyosomes** are cisternae, which are net-like flat membrane-bound cavity structures that make up the Golgi apparatus. The difference between animal and plant cells' dictyosomes is that animal cells' dictyosomes are stacked tightly together, whereas plant cells' dictyosomes are dispersed in the cytoplasm, making them difficult to identify as the Golgi apparatus.

Conceptual Learning